

# Response to Report by Referee 1

“The Hedged Random Forest”

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We thank the referee for the careful reading and constructive suggestions. The report has led to meaningful improvements in the paper. Below we address each point in turn. All section, equation, and table references are to the revised manuscript.

## Main Comments

### Comment 1

*Why not carry out estimation on a second sample using standard OLS with constraints on the weights, which is also a standard approach (Timmermann, 2006)? Ridge would require to tune a parameter, but this could be done by AIC, though ad hoc. Of course, this is a different estimator, but a reader may want to understand the distinction and whether one approach is better than the other.*

**Response.** We have implemented a second-sample benchmark in the spirit of Timmermann (2006). The approach splits the training data in half: the random forest is trained on the first half, and the tree-level weights are estimated on the second half (which the trees have never seen).

A standard OLS estimator on the second sample is not viable here because the number of trees ( $p = 500$ ) exceeds the second-sample size ( $n/2$ ) in some settings, making the system underdetermined. Ridge regression with a sum-to-one constraint handles the high-dimensional regime gracefully; we select the penalty parameter  $\lambda$  by AIC. We refer to this estimator as *2-Step Ridge*.

The empirical results are reported in Figure 1 below (green boxplots). 2-Step Ridge is dominated by HRF at every training-set size we consider, not merely in an isolated instability region: even away from  $n \approx 2p = 1000$ , its mean RMSE ratio relative to HRF sits around 0.87–0.93 (that is, HRF’s RMSE is 7–13% lower). This gap becomes much more severe as the training-set size approaches twice the number of trees ( $n \approx 2p = 1000$ ), because the second-stage regression is then estimated from about as many observations as there are trees to weight; this causes its RMSE to spike relative to HRF by more than an order of magnitude at  $n = 1000$ .

The persistent gap away from  $n = 1000$  is plausibly explained by the sample-splitting procedure itself: the random forest underlying 2-Step Ridge is trained on only the first half of

the training set ( $n/2$  observations), whereas HRF’s forest uses the full sample ( $n$  observations). 2-Step Ridge therefore starts from a structurally weaker forest at every training-set size, quite apart from the instability associated with estimating  $w$  near  $n \approx 2p$ . This, together with the instability documented above, illustrates the benefit of estimating the covariance matrix  $\Sigma$  via nonlinear shrinkage on the full training sample, as HRF does, rather than resorting to sample splitting.

Because this instability is so pronounced, including 2-Step Ridge in the manuscript’s Figure 3 would force an axis scale that obscures the comparison of HRF with the other, better-behaved benchmarks (WRF, CRF, Ridge, and HRF-MinVar), which all lie within a narrow band around a ratio of one. We have therefore decided to keep 2-Step Ridge out of the manuscript itself and report it here instead, where it can be discussed on its own terms.

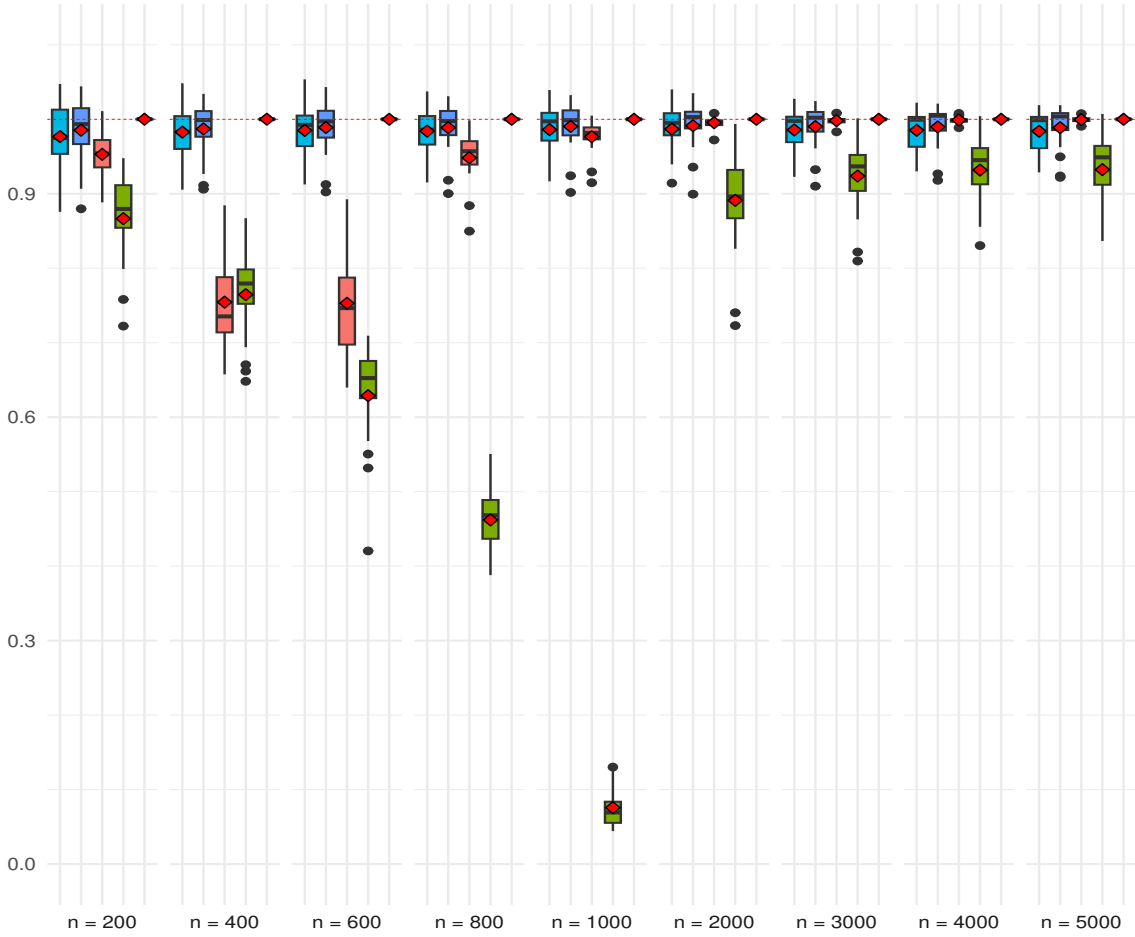


Figure 1: Boxplots of RMSE ratios of HRF relative to WRF (cyan), CRF (blue), Ridge (red), 2-Step Ridge (green), and HRF-MinVar (purple). This is the counterpart to the manuscript’s Figure 3, with 2-Step Ridge added; the wide range needed to accommodate the 2-Step Ridge outliers compresses the detail among the remaining four benchmarks, which is why the manuscript reports only this reduced set. For each training-set size  $n$ , the boxplots are based on the 17 ratios corresponding to the datasets listed in Table 1.

## Comment 2

*Why not carrying out estimation on the same sample using a ridge regression? Again, helping the reader see the distinction and the merits of one method over the other could be useful.*

**Response.** We have added a same-sample ridge regression benchmark (Ridge). As with the second-sample variant, we impose the sum-to-one constraint and select  $\lambda$  by AIC. Because the constraint is an equality, the standard ridge hat matrix must be corrected; we derive the exact effective degrees of freedom analytically via the economy SVD of the tree-prediction matrix.

The results are reported alongside the other benchmarks in Figure 3 (red boxplots) and in Appendix B. Ridge is a reasonably strong benchmark, but it is dominated by HRF, on balance, at every training-set size.

## Comment 3

*Why not focusing on variance minimisation, as for a minimum variance portfolio? Once the weights are derived this way and fixed, the bias, which is a constant, could be removed in a second step, when forecasting.*

**Response.** We have implemented the minimum variance portfolio as an additional benchmark, which we refer to as *HRF-MinVar*. The weights are obtained by minimising  $\mathbf{w}^\top \hat{\Sigma} \mathbf{w}$  subject to  $\mathbf{1}^\top \mathbf{w} = 1$  and the leverage constraint  $\|\mathbf{w}\|_1 \leq \kappa$ . The systematic bias of the combined predictor,  $\hat{b} = \mathbf{w}^\top \hat{\boldsymbol{\mu}}$  where  $\hat{\boldsymbol{\mu}}$  denotes the vector of in-sample tree biases used in the HRF objective, is then added back to each test-set forecast. We compute this benchmark for both the sample and the shrunk (Ledoit–Wolf nonlinear shrinkage) covariance estimator and for  $\kappa \in \{1, 2\}$ , mirroring the HRF design; to keep the comparison with HRF (shrinkage,  $\kappa = 2$ ) as direct as possible, the main text and tables report the matching HRF-MinVar (shrinkage,  $\kappa = 2$ ) variant.

The results are reported in Figure 3 (purple boxplots) and in Appendix B, and are discussed in Comment 4 below.

We would like to stress, however, that we still prefer to keep the bias term  $(w' \hat{\boldsymbol{\mu}})^2$  inside the joint optimization problem (2.5)–(2.7), rather than adopting the two-step MinVar-plus-correction approach as our main method. The reason is that the joint objective gives the weights themselves a chance to hedge against bias: trees with a larger estimated bias are automatically downweighted, in the same way that a mean-variance investor tilts a portfolio away from assets with an unattractive risk-return trade-off. The two-step approach, by contrast, fixes the weights on variance grounds alone and can only correct the resulting aggregate bias *after the fact*, through a single constant shift; it cannot differentially downweight the more biased trees. On our i.i.d. datasets, tree biases turn out to be small and fairly homogeneous, so this distinction happens not to matter much empirically, as the results below confirm. But keeping the bias term in the objective is what gives HRF its built-in insurance against bias in general, and, as we show in our response to Comment 4 below, this insurance does come into play once we move to the time-series application of Section 4.5:

there, dropping the bias term from the optimization measurably erodes HRF’s advantage over RF. We would not want to give up this insurance merely because it is not needed on the i.i.d. datasets studied here.

## Comment 4

*Individual trees tend to have low bias but high variance, so the variance component may dominate in practice. This justifies the use of the sample mean for the estimator of the bias of the individual trees, i.e. they are small numbers close to zero dominated by the variance. Would the estimator perform the same or perhaps better if we just used a minimum variance portfolio approach and forgot about the bias?*

**Response.** The empirical results for the HRF-MinVar benchmark described in Comment 3 provide a direct answer; see Figure 3 and Appendix B. HRF-MinVar (shrinkage,  $\kappa = 2$ ) tracks HRF (shrinkage,  $\kappa = 2$ ) almost exactly, with RMSE ratios between 0.999 and 1.001 at every training-set size, confirming the referee’s intuition that the bias term in the joint mean–variance objective contributes little in practice, at least for i.i.d. data. Individual tree biases are indeed small relative to the variance, so the HRF objective is effectively dominated by the variance component, at least for i.i.d. data.

This conclusion, however, is specific to the i.i.d. setting and does not carry over to genuinely dependent, time-series data. The FRED-MD financial-econometric application added in response to the Associate Editor (Section 4.5) gives us exactly such a testbed, since there  $\hat{\mu}$  and  $\hat{\Sigma}$  are re-estimated on a rolling basis from serially dependent tree residuals, rather than computed once from a single i.i.d. training sample. Repeating the same HRF-MinVar exercise for all 41 FRED-MD series (shrinkage,  $\kappa = 2$ ,  $h = 1$ ) tells a different story: the mean RMSE ratio of HRF-MinVar relative to RF is 1.007, essentially wiping out HRF’s mean RMSE ratio of 0.982 over the same series (Table 3), and HRF-MinVar improves on plain RF for only a handful of the 41 series, compared to HRF’s broad-based improvement. The gap between HRF and HRF-MinVar is largest precisely in the credit/term-spread group, where HRF’s gains over RF are also largest. So for time-series data, unlike for the i.i.d. datasets studied above, dropping the bias term from the optimization is not innocuous: the rolling, time-varying tree biases are apparently large and persistent enough, relative to the variance, that the joint objective’s automatic downweighting of biased trees actually earns its keep.

This finding is informative in three directions. First, it validates the HRF design on i.i.d. data: the method works well even though the bias estimation step is noisy, because the variance component is what matters there. Second, it provides a useful robustness check for practitioners working with i.i.d. data: a pure variance-minimization approach with a post-hoc bias correction is equally effective and even simpler to implement in that setting. Third, and consistent with our response to Comment 3, it shows that this simplification is not a free lunch in general: on time-series data the bias term earns its place in the joint optimization problem (2.5)–(2.7), which is exactly why we keep it there as our default design rather than adopting the two-step MinVar-plus-correction approach.

## Comment 5

*What is very interesting about negative weights is that they should work better in situations where the trees are correlated. From portfolio optimisation theory we know that the weights will be all positive if the correlations of the trees were small. However, negative weights become more likely as correlations increase. Hence, the authors’ method becomes even more important and relevant when features are correlated, making tree decorrelation more difficult. I think remarking on the properties of portfolio weights (especially in light of Point 4) as a function of correlations would be relevant.*

**Response.** We thank the referee for this insightful observation, which we now make explicit in the paper (Section 4.3).

The connection follows directly from classical results in portfolio theory. When the trees are mutually uncorrelated, the minimum-variance weights are all positive: there is no benefit to short positions because each tree contributes independent variance reduction. On the other hand, if some pairwise tree correlations are non-zero, the optimal portfolio can in principle reduce variance further by assigning negative weights to some trees (that is, short-selling trees). The leverage constraint  $\|\mathbf{w}\|_1 \leq \kappa$  controls the degree to which this is permitted:  $\kappa = 1$  restricts weights to be non-negative (no short-selling), whereas  $\kappa = 2$  allows negative weights as long as they are not too large in (the sum of) absolute values.

In the random forest setting this has a natural interpretation. Trees are constructed to be decorrelated via feature subsampling (the `mtry` parameter), but this decorrelation may be far from perfect when the input features are themselves highly correlated. In such datasets, trees may remain substantially correlated, and the gains from  $\kappa = 2$  over  $\kappa = 1$  are expected to be largest. Conversely, when features are nearly independent, and trees are already well-decorrelated,  $\kappa = 1$  generally suffices, and the additional flexibility from  $\kappa = 2$  adds little. This provides a characterisation of when HRF with negative weights is most valuable, and it is consistent with the empirical patterns we observe across the benchmark datasets. The corresponding passage has been added to Section 4.3 of the revised manuscript.

## Additional Comments

### Additional Comment 1

*It would be interesting to know whether one can learn something about the methodology and the HRF estimator from the individual characteristics of the datasets (e.g., variable correlations, fat-tailed data etc.). For instance, are there certain data features that systematically relate to larger performance gains?*

**Response.** We investigated this question across the 17 datasets used in our study (Table 1). For each dataset we computed five characteristics: the average pairwise absolute correlation among the input features; the dispersion in the  $p = 500$  individual trees’ out-of-bag accuracy  $tPE_j$  (eq. (4.4), the same out-of-bag construction used for the WRF benchmark), measured by its coefficient of variation (CV), i.e. the ratio of the standard deviation to the mean of the  $p = 500$  values  $tPE_1, \dots, tPE_p$  within each dataset, as a scale-free measure of how heterogeneous tree *quality* is within that dataset — some trees may simply be more accurate

than others, and HRF can downweight the persistently worse ones; the excess kurtosis of the target variable, as a measure of fat-tailedness; the number of features; and the ratio of the training-set size to the number of features. We related each characteristic (Spearman correlation) to HRF’s performance gain, the RMSE ratio of HRF relative to the equal-weighted RF, averaged across *all* training-set sizes  $n \in \{200, \dots, 5000\}$ .

| Characteristic                                              | $\rho$ | $p$ -value |
|-------------------------------------------------------------|--------|------------|
| Feature correlation                                         | −0.07  | 0.80       |
| Tree-quality dispersion (CV = SD/mean of OOB tree $tPE_j$ ) | −0.72  | 0.002      |
| Target kurtosis                                             | 0.08   | 0.76       |
| Number of features $p$                                      | 0.05   | 0.85       |
| $n/p$                                                       | 0.13   | 0.61       |

Raw feature correlation and target kurtosis show essentially no relationship with HRF’s performance gain across our 17 datasets. Dimensionality ( $p$  and  $n/p$ ) shows a weak, statistically insignificant relationship.

The dispersion in individual-tree quality (Figure 2) is the only characteristic with a clear, statistically significant relationship, and it is, as one might expect, *negative*: datasets in which some trees are persistently much worse than others show the *largest* gains from HRF over the equal-weighted forest. This is consistent with your suspicion that HRF’s ability to downweight worse-performing trees is itself a source of its gains. For instance, *medical\_charges*, *brazilian\_houses*, and *pol* have the most heterogeneous trees in our sample (coefficient of variation  $\approx 0.12$ – $0.15$ ) and HRF/RF RMSE ratios as low as 0.861, whereas *diamonds*, *wine\_quality*, and *MiamiHousing2016* have the most homogeneous trees ( $\approx 0.02$ – $0.03$ ) and ratios close to one (0.97–1.02), averaged across all training-set sizes.

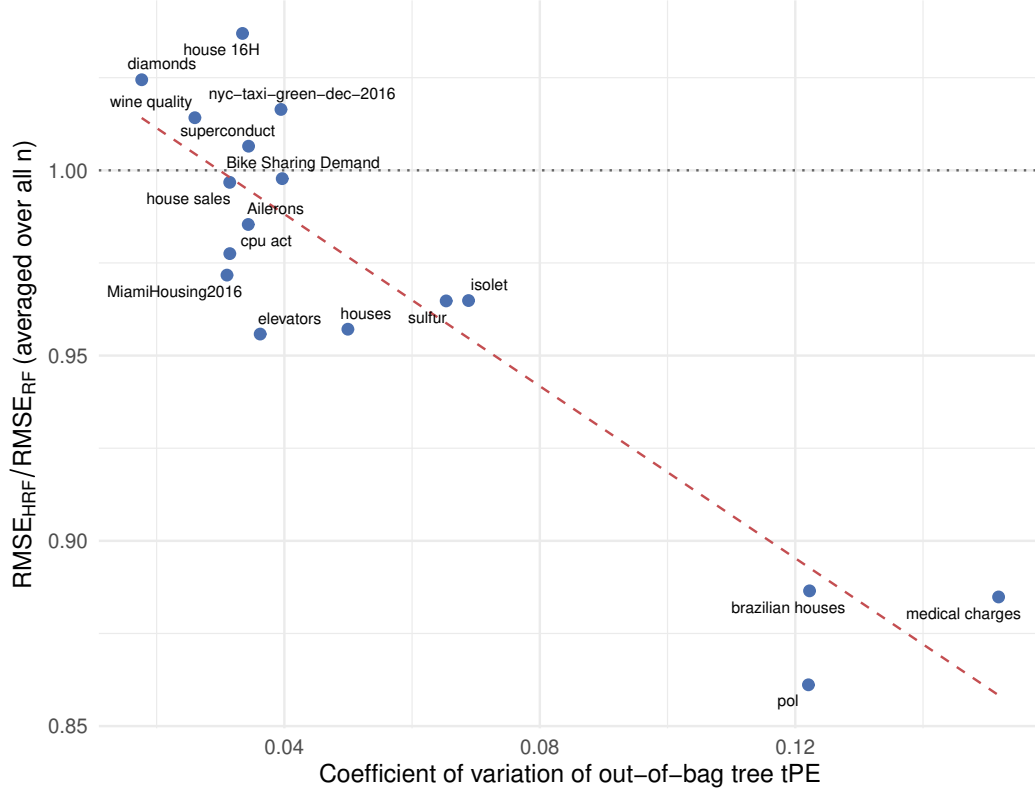


Figure 2: Coefficient of variation of out-of-bag tree accuracy  $tPE_j$  (horizontal axis) against the RMSE ratio of HRF relative to the equal-weighted RF (vertical axis), averaged across all training-set sizes  $n \in \{200, \dots, 5000\}$ , for each of the 17 datasets. The dashed line is a simple linear fit.

We would like to flag this as a genuinely open finding rather than a settled explanation. With only 17 datasets, and only one of five characteristics tested reaching significance, we do not want to over-interpret this correlation, and we have not incorporated it into the manuscript. We report it here in the interest of full transparency; a more systematic investigation of the relationship between tree quality and the value of HRF (for example, using simulated data with a controlled distribution of tree accuracy) would be a natural direction for future work.

## Additional Comment 2

*Many readers will naturally associate the leverage constraint with a Lasso-type penalty and therefore expect a sparse solution, i.e. that only a small number of trees receive nonzero weights. However, the additional portfolio equality constraint (i.e., the requirement that the weights sum to one) changes the geometry of the optimisation problem, and often leads instead to a dense weighting scheme. It might be helpful to include a short remark about this.*

**Response.** We have added Remark 3.4 to Section 3 of the revised manuscript. [RESPONSE PENDING]

## Additional Comment 3

*In section 3, the authors define their specific implementation of the estimator: shrunk covariance matrix and a leverage constraint equal to 2. Later in the paper, they compare results with a leverage constraint equal to one and also compare to the estimator that uses the standard sample covariance matrix. A number of different intuitive notations are used to define these. It would be ideal if this notation or some similar version could be used everywhere. Using shrinkage/sample and  $\kappa$  (the leverage constraint) equal to 1 or 2, everywhere is a possible solution.*

**Response.** We thank the referee for this suggestion. We have unified the notation throughout the paper along the lines proposed. Each variant of the estimator is now identified by two labels: the covariance estimator used (“shrinkage” for the Ledoit–Wolf NLS estimator, “sample” for the plain sample covariance) and the leverage parameter  $\kappa \in \{1, 2\}$ . For example, the main HRF specification discussed in Section 3 is now consistently referred to as HRF (shrinkage,  $\kappa = 2$ ), and this labelling is used uniformly in the text, all tables, and the appendix, including the notation tables in Section 4.2 and the appendix.

## References

Allan Timmermann. Forecast combinations. In G. Elliott, C. W. J. Granger, and A. Timmermann, editors, *Handbook of Economic Forecasting*, pages 135–196. North Holland, Amsterdam, 2006.